

Course Detail and Syllabus

Course Title	Physics		
Course Code		Credit	4
Core/Elective	CEF	Semester	I/II
Prerequisite Knowledge	Basic Knowledge of Physics		

CO	Statement
PH11101.1	The course provides basics of electromagnetism and electrodynamics which are required for Electrical Engineering, Electronics & Communication Engineering, Computer Science & Engineering professionals for development of advance technology.
PH11101.2	Development of basic understanding of quantum mechanics to address the engineering based problems at molecular level, required for Electrical Engineering, Electronics & Communication Engineering, Computer Science & Engineering professionals for the development of advance technologies such as quantum computation and quantum devices.
PH11101.3	The solid state and semiconductor devices are fundamental units of every electronic equipment. The course provides basic understanding of solid state physics to discover efficient electronic devices. It also helps in the development of new materials.

Syllabus

Electrodynamics

Gradient, divergence and curl operations, Spherical and Cylindrical Coordinates. Gauss divergence theorem and Stoke's theorem. Poisson's and Laplace equation, Working of Helmholtz galvanometer. Magnetic vector potential, Displacement current, Maxwell's equations (Integral and differential forms) in free space, Propagation of electromagnetic waves in free space

Quantum Mechanics

Wave particle duality, Wave packets, Phase and group velocity, Heisenberg's uncertainty principle and its application, Wave function and its physical interpretation, Probabilities and Normalization, Time independent and dependent Schrodinger wave equation and its simple applications.

Solid State Physics

Crystal structure, Space lattice, Unit cell, Miller indices, Interplaner spacing, X-ray diffraction and Bragg's law, Diamagnetism, Paramagnetism and Ferromagnetism, Hysteresis curve, Curie-Weiss Law.